

Krishnaa Vadivel | Computational Physics | Distributed Computing | Machine Learning

📞 516-853-5663 • ✉ srikrishnaa.careers@gmail.com • in sri-krishnaa-ece-phd
Website | GitHub: krishnaa423 | LeetCode: krishnaa42342 | Docker Hub: krishnaa42342

Summary

Computational physics researcher building distributed scientific software and using machine learning to accelerate calculations in first-principles materials research. Experience spans Python/C++, CUDA/ROCm, MPI/OpenMP, Linux-based HPC environments, Docker containers for packing applications across environments, and scientific ML.

Research Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Built distributed simulation software in Python and C++ for leadership-class HPC systems, using MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc on Frontier, Perlmutter, and other national-lab clusters.
- Formulated coherent-state and path-integral viewpoints for exciton-phonon dynamics in self-trapped excitons and excitonic polarons, connecting DFT, DFPT, and GW/BSE-style methods.
- Used PyTorch, PyTorch Geometric, and equivariant graph neural networks to accelerate self-trapped-exciton calculations and related materials simulations.
- Packaged scientific applications in Docker containers for deployment across different computing environments and used distributed debuggers such as TotalView and Linaro DDT to track down bugs on HPC systems.
- Co-authored a 2025 *Journal of the American Chemical Society* paper on ultrafast self-trapped exciton formation in lead-free double halide perovskites and presented related APS research in 2024, 2025, and 2026.

Industry Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed CUDA-based software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Built full-stack server applications with C#, ASP.NET, and Microsoft SQL Server for database design, real-time remote tool testing, and calibration of temperature and pressure sensors.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support physics-driven engineering studies.

FindLight

Photonics Technical Writer

2018–2019

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Exciton-Phonon HPC Research: Developed distributed first-principles calculations for self-trapped excitons, many-body theory, and scalable materials simulations.

Equivariant Scientific ML: Developed PyTorch Geometric equivariant models to accelerate molecular and excitonic calculations.

Meep Ring Resonator FDTD: Designed a nanophotonic ring-resonator band-pass filter in telecom bands using Meep-based FDTD simulation.

dolfinx Finite-Element Analysis: Applied dolfinx-based finite-element analysis to frequency studies of steel pipes in well environments.

Education

Yale University <i>PhD, Electrical Engineering</i>	<i>2021–present</i>
Yale University <i>MPhil, Electrical Engineering</i>	<i>2023</i>
Yale University <i>MS, Electrical Engineering</i>	<i>2023</i>
Cornell University <i>MEng, Electrical and Computer Engineering</i>	<i>2017–2018</i>
Cornell University <i>BS, Electrical and Computer Engineering</i>	<i>2013–2017</i>

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, SQL, C#
Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, SciPy, SymPy, matplotlib, dolfinx
Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks
Platforms: Linux, macOS, Windows, Docker, Docker Compose, Kubernetes, gcloud, Frontier, Perlmutter
Domains: First-principles materials modeling, many-body theory, distributed computing, computational physics, numerical PDEs